ContentForge Pro — Production-Ready Content Pipeline With API, Database and Docker

> **Short description:** ContentForge's 5-agent LangGraph writing team, served as a production API: FastAPI with typed requests and safe errors, SQLite run history, request-ID logging, a per-caller rate limit, a Docker image with no secrets inside, API tests with a faked team, and an LLM-judge eval suite that catches prompt regressions.

Tech: Python · FastAPI · SQLite · Docker · pytest · LangGraph · OpenAI API · **Type:** Multi-agent service (Supervisor + 4 specialists, behind an API) · **Level:** Upper-Intermediate

Overview

ContentForge Pro turns a topic into a short, sourced, reviewed article — over HTTP. A client sends `POST /run-task` with a topic; a supervisor-led team of agents (Research, Analysis, Writer, Reviewer) writes the article; the API returns it with whether the Reviewer approved it and why the run stopped.

Around the agents sits the production layer: typed request and reply shapes, safe 4xx/5xx errors, every run saved to SQLite and findable by ID, one request ID on every log line, a rate limit per caller, a Docker image that takes its key only at run time, and two kinds of tests — fast API tests with the team faked, and an eval suite that scores the real articles.

Why

A multi-agent system that only runs from a terminal can't be called by a web page, another service or a teammate. Without saved runs, nobody can answer "what went wrong on that request?". Without tests that score the output, a prompt edit that makes every article a little worse ships silently. ContentForge Pro adds exactly those missing pieces — without changing a single agent.

Features

- **REST API** (FastAPI): `POST /run-task`, `GET /runs/{run_id}`, `GET /health`, with interactive docs at `/docs`.
- **Typed contract** — Pydantic request/reply models; a bad body is a 422 before any model is called.

- **Safe errors** — 400 for a blank topic, 404 for an unknown run, 429 over the rate limit, and a plain 500 for a crash (the traceback stays in the server log).
- **Run history** — every run saved to SQLite as `started` → `completed` / `failed`, with its input, article and routing log.
- **Request IDs** — one short ID on every log line of a request, and in the `X-Request-ID` reply header.
- **Rate limiting** — a rolling 60-second window per caller.
- **Docker** — the key is passed only at `docker run`; `.dockerignore` keeps `.env` and local data out of the image; the app refuses to start without its key.
- **Tests at every layer** — unit tests, API tests with the team faked (no model calls), live integration tests, and a 10-task eval suite with an LLM judge and a regression check.

Demo

```
$ curl -X POST localhost:8000/run-task \
    -H "Content-Type: application/json" \
    -d '{"topic": "electric bikes"}'
{"run_id":1,"status":"completed","accepted":true,
 "reason":"reviewer accepted the draft",
 "article":"E-Bikes: How Far, How Fast ... [battery-review] ..."}

$ curl localhost:8000/runs/1
{"id":1,"input":"electric bikes","output":"E-Bikes: How Far ...",
 "log":"research_agent: no research yet\n...\nEND: reviewer accepted
 the draft","status":"completed","created_at":"2026-01-01T00:00:00+00:00"}

$ curl -X POST localhost:8000/run-task \
    -H "Content-Type: application/json" \
    -d '{"topic": "deep sea mining"}'
{"run_id":2,"status":"completed","accepted":false,
 "reason":"research found no facts for this topic","article":""}
```

(Replies are really one line each — shown wrapped. The article varies from run to run.)

Architecture

```
client ── POST /run-task {"topic": ...}
           │
           ▼
   api/main.py    request-ID middleware (X-Request-ID)
   api/routes.py  rate limit per caller ── over it ──► 429
```

```
                  blank topic ──────────────────────────► 400
            │
            ▼
   db/database.py INSERT run, status "started"
            │
            ▼
   pipeline.py    run_pipeline(topic)
                  Supervisor → Research → Analysis
                             → Writer ⇄ Reviewer (max 3 revisions)
            │
            ├─ crash ──► UPDATE "failed" ──► plain 500
            ▼
   db/database.py UPDATE output, log, status "completed"
            │
            ▼
   200 {run_id, status, accepted, reason, article}

GET /runs/{run_id} ──► the saved run, or 404
evals/  10 fixed topics ──► run_pipeline ──► code check or judge ──► score
```

- **One door into the team** — the API and the eval suite both call `run_pipeline()`; tests replace that one function with a fake.
- **Three run states** — the `started` row is saved before the team runs, so a crashed run still leaves a record.
- **The agents are unchanged** — routing, revision limit and prompts are the original ContentForge team's.

Project Structure

```
<repo folder>/   (ContentForge-Pro-...-API-Database-And-Docker/)
├── README.md               the repo's front page
├── LICENSE                 lets others reuse your code (MIT)
├── .gitignore              keeps .env, .venv/, runs.db out of git
├── .env.example            the setting names, no real values
├── .env                    your real key — never committed
├── pyproject.toml          makes src/contentforge an installable package
├── requirements.txt        the libraries, for pip install -r
├── Dockerfile              builds the API's container image
├── .dockerignore           keeps .env and runs.db out of the image
├── src/
│   └── contentforge/
│       ├── __init__.py
│       ├── main.py             asks for a topic, prints the report
│       ├── config.py           Config, load_config()
```

```
│       ├── exceptions.py       MissingConfigError
│       ├── schemas.py          ReviewVerdict (score, feedback)
│       ├── state.py            TeamState, start_state()
│       ├── graph.py            build_graph(): 5 nodes, 1 edge
│       ├── pipeline.py         run_pipeline(): the one door in
│       ├── models/
│       │   ├── __init__.py
│       │   └── llm.py          get_model()
│       ├── agents/
│       │   ├── __init__.py
│       │   ├── supervisor.py      decide_next(), supervisor()
│       │   ├── research_agent.py  run_research(), research_node
│       │   ├── analysis_agent.py  run_analysis(), analysis_node
│       │   ├── writer_agent.py    run_writer(), writer_node
│       │   └── reviewer_agent.py  run_reviewer(), reviewer_node
│       ├── tools/
│       │   ├── __init__.py
│       │   └── lookup.py       lookup_facts (@tool)
│       ├── prompts/
│       │   ├── research.txt
│       │   ├── analysis.txt
│       │   ├── writer.txt
│       │   └── reviewer.txt
│       ├── utils/
│       │   ├── __init__.py
│       │   ├── logger.py       get_logger()
│       │   └── prompts.py      load_prompt(name)
│       ├── api/
│       │   ├── __init__.py
│       │   ├── main.py         the app uvicorn starts
│       │   ├── routes.py       /health, /run-task, /runs/{run_id}
│       │   └── schemas.py      TaskRequest, TaskResponse
│       ├── db/
│       │   ├── __init__.py
│       │   ├── database.py     save and fetch runs (SQLite)
│       │   └── models.py       Run: one saved run's shape
│       └── evals/
│           ├── __init__.py
│           ├── tasks.py        10 fixed topics and checks
│           ├── judge.py        score_with_judge()
│           ├── run_eval.py     runs the suite, saves report.json
│           └── regression_check.py  breaks a prompt, checks the drop
├── tests/
│   ├── unit/
│   │   ├── test_lookup.py      no API key needed
│   │   └── test_supervisor.py  no API key needed
│   ├── api/
```

```
│   │   └── test_routes.py      the API, with the team faked
│   └── integration/
│       ├── test_live_agents.py real API calls
│       └── test_judge.py       real API calls
└── data/
    └── sample/
        └── facts.json          the facts Research can look up
```

Getting Started

Requirements: Python 3.10+, an OpenAI API key, and Docker (for the container only).

```
git clone <this-repo-url>
# the \ joins the two lines into one folder name
cd ContentForge-Pro-Production-Ready-Content-Pipeline-With-API-\
Database-And-Docker
python -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate
pip install -r requirements.txt
pip install -e .
cp .env.example .env               # then put your key in .env
```

Configuration

Variable	Required	Default	Meaning
`OPENAI_API_KEY`	yes	—	Your OpenAI API key; the app will not start without it
`LOG_LEVEL`	no	`INFO`	`DEBUG`, `INFO`, `WARNING` or `ERROR`

Setting	Where	Default
Chat model	`models/llm.py` → `MODEL_NAME`	`gpt-4o-mini`
Pass mark	`agents/reviewer_agent.py` → `PASS_SCORE`	`7` (out of 10)
Revision limit	`agents/supervisor.py` → `MAX_REVISIONS`	`3`
Topic length	`api/schemas.py` → `TaskRequest`	1-200 characters
Rate limit	`api/routes.py` → `RATE_LIMIT`, `RATE_WINDOW_SECONDS`	5 requests per 60 s, per caller

Setting	Where	Default
Database file	`db/database.py` → `DB_PATH`	`runs.db`
Eval pass bar	`evals/regression_check.py` → `PASS_BAR`	`0.8`
Known facts	`data/sample/facts.json`	electric bikes, remote work, solar panels

Usage

Run from the repo root:

```
uvicorn contentforge.api.main:app --reload     # the API on port 8000
python -m contentforge.main                    # the terminal app
```

Open `http://localhost:8000/docs` to try every route in the browser.

Route	What it does
`GET /health`	`{"status":"ok"}`
`POST /run-task`	Body `{"topic": "..."}` ; returns `run_id` , `status` , `accepted` , `reason` , `article`
`GET /runs/{run_id}`	The saved run: `input` , `output` , `log` , `status` , `created_at` ; 404 if unknown

In a container:

```
docker build -t contentforge-pro .
docker run -p 8000:8000 --env-file .env contentforge-pro
curl localhost:8000/health
```

Running the Tests

Run from the repo root:

```
# free — no model calls
python tests/unit/test_lookup.py
python tests/unit/test_supervisor.py
pytest tests/api -v
```

```
# real API calls — needs a valid OPENAI_API_KEY
python tests/integration/test_live_agents.py
pytest tests/integration/test_judge.py -v
```

The unit tests (plain scripts, the same as in the original team project) check the lookup tool and every routing rule. The API tests check every route and status code — 200, 400, 404, 422, 429 and a plain 500 — and the saved rows, with the team replaced by a fake. The integration tests run the real team and check the judge on answers with a known verdict.

Article quality (real API calls):

```
python -m contentforge.evals.run_eval           # score + report.json
python -m contentforge.evals.regression_check   # prove a bad change drops it
```

`run_eval` runs 10 fixed topics — exact `[source]` checks, judge rules and unknown topics — and prints the score. `regression_check` runs it, pauses while you remove the `[source]` rule from Research's prompt, runs it again, and reports whether the score fell below the bar.

Error Handling

Situation	What happens
`OPENAI_API_KEY` missing	The app (or container) stops at startup with `MissingConfigError`
Missing, empty or over-long `topic`	422, nothing saved
Topic of only spaces	400 `topic cannot be empty`
Too many requests from one caller	429 `Rate limit exceeded`, before any model call
Crash inside the team (bad key, timeout)	Plain 500; full traceback in the server log with the request ID; run saved as `failed`
Unknown run ID	404 `run not found`
Topic with no known facts	200, `accepted: false`, empty article, and the reason
Writer and Reviewer never agree	200, `accepted: false`, the last draft, reason `revision limit (3) reached`

Design Decisions

- **One door into the team** (`run_pipeline`) — the API never reaches into the graph's state, and tests fake one function.
- **Save before running** — a `started` row in its own transaction, so a crash still leaves an honest `failed` record.
- **A plain `def` route** — the agents' model calls block, so FastAPI runs the route in its thread pool and other requests keep moving.
- **Secrets only at run time** — nothing secret in any image layer; the app checks for its key at startup.
- **Test layers** — free unit and API tests on every change; the paid eval suite before a prompt or model change.
- **Rules written before running** — the eval tasks describe a good article, not whatever the team produced.
- **A suite that has been seen to fail** — the regression check proves a real prompt break drops the score.

Limitations and Roadmap

- The rate limit lives in memory: it resets on restart, and each copy of the app counts on its own. A shared store (Redis) fixes that.
- SQLite is one file inside the container — lost when the container is removed, and not shared between copies. Mount a volume, or move to Postgres.
- A request waits for the whole team (several seconds). A job queue with `GET /runs/{run_id}` polling fits longer runs.
- No login or API keys for callers.
- Possible next steps: caller API keys, a job queue, Postgres, a CI pipeline that runs the free tests on every push, and cost tracking per run.

License

MIT — see `LICENSE` .